

Clustering in Machine Learning

- Unsupervised Learning & K-Means
- Course: Data Science

What is Clustering?

- Grouping similar data points
- Maximize similarity within cluster

Characteristics

- No labels
- Similarity-based
- Output: clusters

Applications

- Market segmentation
- Bug grouping
- Recommendation systems

Types of Clustering

- Partition (K-Means)
- Hierarchical
- Density-based
- Model-based

K-Means Overview

- Divides data into K clusters
- Each has a centroid

Algorithm Steps

- 1. Choose K
- 2. Init centroids
- 3. Assign points
- 4. Update centroids
- 5. Repeat

Distance Formula

- Euclidean Distance
- $\text{sqrt}((x_2-x_1)^2 + (y_2-y_1)^2)$

Dataset Example

- Points: A(2,10), B(2,5), C(8,4), D(5,8), E(7,5), F(6,4)

Initialization

- $C1 = A(2, 10)$
- $C2 = C(8, 4)$

Cluster Assignment

- Cluster1: A,B,D
- Cluster2: C,E,F

Centroid Update

- $C1=(3,7.67)$
- $C2=(7,4.33)$

Final Clusters

- Cluster1: A,B,D
- Cluster2: C,E,F

Advantages

- Simple
- Fast
- Scalable

Limitations

- Choose K
- Sensitive to init
- Outliers affect

Elbow Method

- Plot K vs WCSS
- Choose elbow point

Python Demo

- Use sklearn KMeans
- Fit and plot clusters

Summary

- Clustering groups unlabeled data
- K-Means is widely used

Questions

- Where used in SE?
- Effect of K ?